Matthew Kuney

Kennett Square, PA
484-995-3854 kuney.matthew@gmail.com linkedin.com/in/matthewkuney matthewkuney.github.io

Education

The Pennsylvania State University **Expected May 2030**
Bachelor of Science in Mechanical Engineering *University Park, PA*

Relevant Coursework

- Ordinary and Partial Differential Equations
- Matrices
- ME 101: Toy Fundamentals

Experience

Penn State Learning Factory **Sep 2026 – Present**
Learning Factory Teaching Assistant *University Park, PA*

- Operate and maintain equipment across five fabrication labs: wood shop, metal shop, laser engraving, additive manufacturing, and electronics/IoT.
- Most active in the additive manufacturing, laser engraving, and woodworking labs, with hands-on welding and soldering experience.
- Lead hands-on safety and equipment trainings for groups of 12+ students, preparing them to independently and safely use Learning Factory machinery.
- Respond to shop emergencies and enforce safety protocols across multiple specialized fabrication labs.

Progressive Pool Management Inc. **May 2024 – Aug 2025**
Head Lifeguard *Garnet Valley, PA*

- Managed independent 10-hour shifts, including facility opening and closing procedures and daily operations.
- Maintained proper pool chemical levels through routine water testing and chemical level logs.
- Responded to emergencies and enforced facility safety policies and procedures.

Projects

Pick-and-Place Robot | *Arduino, CAD, Sheet Metal Fabrication, 3D Printing*

- Designed and built a drone-controller-operated pick-and-place robot from scratch, including a welded sheet-metal chassis, drive motors, and a servo-actuated arm.
- Modeled and 3D-printed a custom gripper mechanism to grasp and relocate objects.
- Integrated an Arduino microcontroller with a buck converter and battery system to power and control the robot's motors and electronics.

Fully Mechanical 3D-Printed Excavator Model | *Onshape CAD, 3D Printing*

- Designed and fabricated a fully mechanical, non-electronic excavator model with 25+ parts in Onshape, driven entirely by springs, gears, nuts, and bolts.
- Engineered a 3-degree-of-freedom arm and bucket assembly, with the cab swiveling on a bevel gear drivetrain.

Technical Skills & Certifications

CAD & Design: Fusion 360, Onshape
Fabrication: 3D Printing, Laser Cutting, Metal Fabrication, Woodworking, Welding
Electronics: Arduino, Circuit Design
Other: Mechanical Design, Prototyping
Certifications: Penn State Learning Factory Safety & Equipment Training (2026), American Heart Association First Aid (2026)

Leadership / Extracurricular

ASME (American Society of Mechanical Engineers) **2026 – Present**
Member *University Park, PA*

- Helped organize and run Penn State ASME's Battle Bots competition.
- Hovercraft Competition Team (in progress): designed mounts for electronic and motor components.
- Student Design Challenge Robot Team (in progress): designed a mock competition course based on the rulebook for practice and reference.